Appendix A v1.5: Positivity-Preserving Phase-Space Confinement, Compact Trapping Box Theorem, and Conditional Holographic Closure

Supplementary Artifact v1.5 (Upgraded Edition) — Additive to *Physics v1.0* and Appendices A v1.1–v1.4.1

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Abstract

Appendices A v1.1 through v1.4.1 established that the three-variable phenomenological dynamical system of *Physics v1.0* admits a unique interior equilibrium whose sole linear instability is a subcritical Hopf bifurcation ($l_1 > 0$) across the interior positive parameter orthant, and demonstrated that post-Hopf trajectories ($e > e_c$) escape the physical positive orthant $\mathbb{R}^3_+$ in finite time. Implementing Option 2 of Appendix A v1.1, this v1.5 supplement introduces a modified, smooth (C^∞) soft-wall dynamical system that eliminates finite-time escape while converging exponentially to the v1.0–v1.4 system in the interior as the boundary regularization widths $\epsilon_H, \epsilon_I \to 0^+$. First, we state a *conditional* holographic dictionary for Gate 0 via the covariant Hubeny–Rangamani–Takayanagi (HRT) and Bousso causal-diamond bounds; for general non-geometric Hilbert-space factorizations, Gate 0 remains open. Second, by gating the irreversible-erasure sink with $\Psi_{\epsilon_H}(H_{\text{obs}}) = \tanh(H_{\text{obs}}/\epsilon_H)$ and the derivative-feedback channel with $\Phi_{\epsilon_I}(\dot{I}) = \tanh(\dot{I}/\epsilon_I)$, we prove via Nagumo's subtangentiality theorem that both $\mathbb{R}^3_+$ and the compact box $\mathcal{B} = [0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \dot{I}_{\max}]$ are globally forward-invariant for all $t \geq 0$ (an analytical containment theorem), while multi-initial-condition Poincaré return-map and transverse Floquet multiplier analyses ($\rho(DP) \approx 8.96 \times 10^{-6} \ll 1$) demonstrate numerically that post-Hopf trajectories settle onto an attracting periodic relaxation orbit inside $\mathcal{B}$. Third, because $dH_{\text{obs}}/dt|_* = 0$ at steady state, the first derivative of $\Phi_{\epsilon_I}(\dot{I})$ vanishes identically in the equilibrium Jacobian, leaving $(\gamma^*, \dot{I}^*)$ exact and preserving H^*_{obs}, J, e_c, and l_1 up to ϵ-dependent exponential corrections scaling as $1 - \tanh(\dot{I}^*/\epsilon_I) \sim 2e^{-2\dot{I}^*/\epsilon_I}$ and $\text{sech}^2(H^*_{\text{obs}}/\epsilon_H) \sim 4e^{-2H^*_{\text{obs}}/\epsilon_H}$. Across a parametric sweep $\epsilon_H = \epsilon_I = \epsilon \in [0.005, 0.10]$ on the v1.4 audit slice, $\|J_{\text{v1.5}} - J_{\text{v1.4}}\|_F$ and $|\Delta l_1|$ decay from 2.21×10^{-8} and 1.75×10^{-6} at $\epsilon = 0.10$, to 1.48×10^{-86} and 1.11×10^{-82} at $\epsilon = 0.01$ (where $\text{sech}^2(H^*_{\text{obs}}/\epsilon_H) = 5.38 \times 10^{-695}$), and 2.05×10^{-173} and 6.11×10^{-169} at $\epsilon = 0.005$. All six face conditions of $\mathcal{B}$ are formalized in Lean 4 (`boundary_verification.lean`), and all symbolic equilibria, Taylor tensors, ϵ-scaling laws, and Floquet multipliers are verified via `SymPy`, 60/750/1450-digit `mpmath`, and `SciPy` (`verification_patch_v1_5.py`).

1 Lineage Provenance and Epistemic Scope

Under the immutability rules of the *Physics v1.0* research lineage ([10.17605/OSF.IO/N3DQU]), the original unregularized ODE ansatz analyzed in Appendix A v1.1 and the exact symbolic positivity certificate of Appendix A v1.4/v1.4.1 (10.17605/OSF.IO/U7M6H, 10.17605/OSF.IO/S7WHX) remain frozen and unmodified. Specifically, Appendix A v1.1 identified two minimal modelling options to address the finite-time escape to $\gamma < 0$ across the Hopf boundary $e = e_c$:

1. Restrict the admissible parameter domain of the v1.0 ODE to the linearly stable regime $e < e_c$ with the erasure admissibility bound $\dot{I}_{\text{erase}} \leq c\gamma^* H_{\max}$ enforced; or
2. Define a modified, positivity-preserving dynamical system (v1.5) in which $\mathbb{R}^3_+ = \{(\gamma, H_{\text{obs}}, \dot{I}) \in \mathbb{R}^3 \mid \gamma \geq 0,\ H_{\text{obs}} \geq 0,\ \dot{I} \geq 0\}$ is globally forward-invariant while converging exponentially to the v1.0–v1.4 equilibrium and Hopf scaling in the interior regime ($\epsilon_H, \epsilon_I \ll 1$).

This document executes Option 2 as a distinct versioned dynamical model (v1.5) and formulates a conditional holographic dictionary for the saturation capacity $H_{\max}$.

2 Conditional Holographic Dictionary for Gate 0

In *Physics v1.0*, the observer von Neumann entropy $H_{\text{obs}}(t) = -\text{Tr}(\rho_{\text{obs}} \log_2 \rho_{\text{obs}})$ is bounded by an informational saturation capacity $H_{\max}$ (in bits). The Gate 0 Technical Report recorded the bridge between

an abstract subsystem factorization $\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{obs}} \otimes \mathcal{H}_{\text{env}}$ and geometric horizon entropy $A/(4\ell_P^2 \ln 2)$ as **UNRESOLVED IN GENERALITY** because an arbitrary algebraic tensor factorization need not correspond to a spatially localized region with a well-defined boundary area.

To make the microphysical assumption required to close Gate 0 mathematically explicit, we formulate a conditional covariant causal-diamond localization postulate:

Proposition 1 (Conditional Holographic Closure of Gate 0)**.** *Let $(\mathcal{M}, g_{\mu\nu})$ be a $(3+1)$-dimensional semiclassical spacetime, and suppose the observer subsystem $\mathcal{H}_{obs}$ is defined by the algebra of quantum fields localized within the causal diamond $\mathcal{D}(\mathcal{A})$ of a spacelike ball $\mathcal{A}$ with smooth codimension-two boundary $\partial\mathcal{A}$. Under the covariant Bousso entropy bound and the Hubeny–Rangamani–Takayanagi (HRT) extremal surface prescription, if Σ_{HRT} is the unique minimal-area extremal surface homologous to $\mathcal{A}$ with $\partial\Sigma_{HRT} = \partial\mathcal{A}$ and stationary area $\text{Area}(\Sigma_{HRT}) = A_\Sigma$, then the state-space supremum of the observer's fine-grained von Neumann entropy in bits is fixed by*

$$H_{\max} \equiv \sup_{\rho_{obs} \in \mathcal{S}(\mathcal{D}(\mathcal{A}))} [-\operatorname{Tr}(\rho_{obs} \log_2 \rho_{obs})] = \frac{\text{Area}(\Sigma_{HRT})}{4\ell_P^2 \ln 2} \quad [bits]. \tag{1}$$

Remark 2 (Distinction Between Conditional Closure and General Derivation). Equation (1) closes Gate 0 *conditionally* within the class of spatially localized causal-diamond subsystems admitting a semiclassical holographic or covariant entropy bound. For abstract, non-geometric Hilbert-space factorizations $\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{obs}} \otimes \mathcal{H}_{\text{env}}$ without a spatial boundary $\partial\mathcal{A}$, Gate 0 remains open in generality. None of the dynamical-sector theorems in Sections 3–6 depend on Eq. (1); they hold for any constant $H_{\max} > 0$.

3 Upgraded Soft-Wall Dynamical System and Global Trapping Box Theorem

In the unregularized v1.0 system, finite-time escape from $\mathbb{R}^3_+$ occurs through a two-stage boundary breach: (i) when H_{obs} approaches 0, the constant erasure term $-\dot{I}_{\text{erase}}$ remains active and drives $H_{\text{obs}} < 0$; and (ii) during large-amplitude oscillations above the subcritical Hopf threshold ($e > e_c$), negative entropy rates $dH_{\text{obs}}/dt < 0$ enter the derivative-feedback term $e\,(dH_{\text{obs}}/dt)$ unattenuated at $\dot{I} = 0$, driving $\dot{I} < 0$ and subsequently pulling $\gamma < 0$ via $a\dot{I}\mathcal{E}_{\text{avail}}(1 - \gamma/\gamma_{\max})$.

Note that replacing $-\dot{I}_{\text{erase}}$ with $-\frac{1}{2}\dot{I}_{\text{erase}}[1 + \tanh(H_{\text{obs}}/\epsilon_H)]$ or leaving $e\,(dH_{\text{obs}}/dt)$ ungated at $\dot{I} = 0$ does *not* render $\mathbb{R}^3_+$ forward-invariant, because $\frac{1}{2}[1 + \tanh(0)] = \frac{1}{2} \neq 0$ leaves a negative sink $-\frac{1}{2}\dot{I}_{\text{erase}}$ at $H_{\text{obs}} = 0$ whenever $\gamma < \dot{I}_{\text{erase}}/(2cH_{\max})$, and an ungated $e\,(dH_{\text{obs}}/dt)$ drives $\frac{d\dot{I}}{dt}\Big|_{\dot{I}=0} < 0$ whenever $e\,(dH_{\text{obs}}/dt) < -f\mathcal{P}_{\text{input}}$.

To construct a modified C^∞ dynamical system in which $\mathbb{R}^3_+$ and a compact box $\mathcal{B}$ are unconditionally forward-invariant, let $\epsilon_H, \epsilon_I > 0$ denote boundary regularization widths ($\epsilon_H \ll H_{\max}$, $\epsilon_I \ll \dot{I}^*$) and define the smooth gates

$$\Psi_{\epsilon_H}(H_{\text{obs}}) \equiv \tanh\left(\frac{H_{\text{obs}}}{\epsilon_H}\right), \qquad \Phi_{\epsilon_I}(\dot{I}) \equiv \tanh\left(\frac{\dot{I}}{\epsilon_I}\right). \tag{2}$$

The v1.5 soft-wall dynamical system is defined by

$$\frac{d\gamma}{dt} = a\dot{I}\mathcal{E}_{\text{avail}}\left(1 - \frac{\gamma}{\gamma_{\max}}\right) - b\gamma, \tag{3}$$

$$\frac{dH_{\text{obs}}}{dt} = c\gamma(H_{\max} - H_{\text{obs}}) - \dot{I}_{\text{erase}} \tanh\left(\frac{H_{\text{obs}}}{\epsilon_H}\right), \tag{4}$$

$$\frac{d\dot{I}}{dt} = e \tanh\left(\frac{\dot{I}}{\epsilon_I}\right)\left(\frac{dH_{\text{obs}}}{dt}\right) + f\mathcal{P}_{\text{input}} - g\dot{I}, \tag{5}$$

where all parameters $a, b, c, e, f, g, \mathcal{E}_{\text{avail}}, \mathcal{P}_{\text{input}}, \dot{I}_{\text{erase}}, \gamma_{\max}, H_{\max}, \epsilon_H, \epsilon_I$ are strictly positive.

Theorem 3 (Global Forward-Invariance of $\mathbb{R}^3_+$ and Compact Trapping Box $\mathcal{B}$)**.** *For all strictly positive parameters and every feedback gain $e > 0$:*

1. *The non-negative orthant $\mathbb{R}^3_+ = \{(\gamma, H_{obs}, \dot{I}) \in \mathbb{R}^3 \mid \gamma \geq 0,\ H_{obs} \geq 0,\ \dot{I} \geq 0\}$ is positively invariant under the flow of Eqs. (3)–(5).*

2. *Define the maximum entropy-rate bound* $M_H \equiv c\gamma_{\max} H_{\max} + \dot{I}_{erase}$ *and the processing ceiling*

$$\dot{I}_{\max} \equiv \frac{eM_H + f\mathcal{P}_{input}}{g} > 0. \tag{6}$$

Then the compact box $\mathcal{B} \equiv [0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \dot{I}_{\max}] \subset \mathbb{R}^3_+$ *is positively invariant and globally absorbing within* $[0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \infty)$*. Consequently, every trajectory originating in* $\mathcal{B}$ *exists globally for all* $t \in [0, \infty)$ *and never escapes to negative states or blows up in finite time.*

Proof. Since the vector field $F = (d\gamma/dt, dH_{\text{obs}}/dt, d\dot{I}/dt)$ of Eqs. (3)–(5) is C^∞ on $\mathbb{R}^3$, by Nagumo's subtangentiality theorem a closed convex box $\mathcal{B} = \prod_{i=1}^3 [x_{i,\min}, x_{i,\max}]$ is positively invariant if and only if $F_i(x) \geq 0$ whenever $x \in \mathcal{B}$ with $x_i = x_{i,\min}$ and $F_i(x) \leq 0$ whenever $x \in \mathcal{B}$ with $x_i = x_{i,\max}$. We check all six boundary faces of $\mathcal{B}$:

1. **Face 1 — Lower γ-face ($\gamma = 0$, $\dot{I} \geq 0$):** Evaluating Eq. (3) at $\gamma = 0$ yields

$$\left.\frac{d\gamma}{dt}\right|_{\gamma=0} = a\dot{I}\mathcal{E}_{\text{avail}}(1 - 0) - b(0) = a\dot{I}\mathcal{E}_{\text{avail}} \geq 0, \tag{7}$$

with strict positivity whenever $\dot{I} > 0$.

2. **Face 2 — Upper γ-ceiling ($\gamma = \gamma_{\max}$, $\dot{I} \geq 0$):** Evaluating Eq. (3) at $\gamma = \gamma_{\max}$ yields

$$\left.\frac{d\gamma}{dt}\right|_{\gamma=\gamma_{\max}} = a\dot{I}\mathcal{E}_{\text{avail}}\left(1 - \frac{\gamma_{\max}}{\gamma_{\max}}\right) - b\gamma_{\max} = -b\gamma_{\max} < 0. \tag{8}$$

3. **Face 3 — Lower H_{obs}-face ($H_{\text{obs}} = 0$, $\gamma \geq 0$):** Because $\tanh(0/\epsilon_H) = \tanh(0) = 0$, evaluating Eq. (4) at $H_{\text{obs}} = 0$ gives

$$\left.\frac{dH_{\text{obs}}}{dt}\right|_{H_{\text{obs}}=0} = c\gamma(H_{\max} - 0) - \dot{I}_{\text{erase}} \tanh(0) = c\gamma H_{\max} \geq 0 \quad \forall \gamma \geq 0. \tag{9}$$

4. **Face 4 — Upper H_{obs}-ceiling ($H_{\text{obs}} = H_{\max}$, $\gamma \geq 0$):** Since $H_{\max}/\epsilon_H > 0 \implies \tanh(H_{\max}/\epsilon_H) > 0$,

$$\left.\frac{dH_{\text{obs}}}{dt}\right|_{H_{\text{obs}}=H_{\max}} = c\gamma(0) - \dot{I}_{\text{erase}} \tanh\left(\frac{H_{\max}}{\epsilon_H}\right) < 0. \tag{10}$$

5. **Face 5 — Lower $\dot{I}$-face ($\dot{I} = 0$, arbitrary $\gamma, H_{\text{obs}} \in \mathbb{R}$):** Because $\tanh(0/\epsilon_I) = 0$, both the feedback term $e \tanh(0)(dH_{\text{obs}}/dt)$ and the linear damping $-g(0)$ vanish identically at $\dot{I} = 0$:

$$\left.\frac{d\dot{I}}{dt}\right|_{\dot{I}=0} = e \cdot 0 \cdot \frac{dH_{\text{obs}}}{dt} + f\mathcal{P}_{\text{input}} - g(0) = f\mathcal{P}_{\text{input}} > 0. \tag{11}$$

6. **Face 6 — Upper $\dot{I}$-ceiling ($\dot{I} = \dot{I}_{\max}$ on $\mathcal{B}$):** For any $(\gamma, H_{\text{obs}}) \in [0, \gamma_{\max}] \times [0, H_{\max}]$, $dH_{\text{obs}}/dt \leq c\gamma_{\max} H_{\max} \leq c\gamma_{\max} H_{\max} + \dot{I}_{\text{erase}} = M_H$. Since $0 \leq \tanh(\dot{I}/\epsilon_I) \leq 1$, at $\dot{I} = \dot{I}_{\max}$ we have

$$\left.\frac{d\dot{I}}{dt}\right|_{\dot{I}=\dot{I}_{\max}} \leq eM_H + f\mathcal{P}_{\text{input}} - g\dot{I}_{\max} = 0. \tag{12}$$

Since F points inward on every boundary face of $\mathcal{B}$ and $\mathcal{B}$ is compact, standard ODE global existence guarantees that all trajectories in $\mathcal{B}$ are defined and confined to $\mathcal{B}$ for all $t \geq 0$. □

Analytical Containment vs. Numerical Relaxation Limit Cycle: Theorem 3 is a global analytical containment theorem proving that no trajectory can exit $\mathcal{B}$. In the post-Hopf regime ($e > e_c$) where the unique interior equilibrium is unstable, multi-initial-condition Poincaré return-map integration and linearized transverse Floquet multiplier evaluation ($\rho(DP) = 8.96 \times 10^{-6} \ll 1$, Section 6, Table 3) provide direct numerical evidence that trajectories repelled from the equilibrium converge onto an attracting periodic relaxation oscillation inside $\mathcal{B}$ whose closest approach to $\gamma = 0$ and $H_{\text{obs}} = 0$ is governed by ϵ_H, ϵ_I.

4 Formal Kernel Verification in Lean 4 (All Six Faces of $\mathcal{B}$)

The accompanying Lean 4 module `boundary_verification.lean` formalizes Eqs. (3)–(6) over Mathlib's $\mathbb{R}$ and proves all six boundary face conditions, the auxiliary bound `dHobs_dt_le_M_H`, and the composite 6-face certificate `compact_box_nagumo_certificate` using explicit Mathlib monotonicity lemmas (`mul_le_mul_of_nonneg_left`, `mul_le_mul_of_nonneg_right`, `mul_assoc`, `mul_div_cancel₀`, and `linarith`) with zero `sorry` axioms and zero `nlinarith` heuristics:

```
/-- Upper ceiling theorems (Faces 2, 4, 6) and lower face theorems (Faces 1, 3, 5). -/
theorem dHobs_dt_neg_at_ceiling
    (c H_max I_erase epsH : ℝ) (hIe : 0 < I_erase) (htanh : 0 < Real.tanh (H_max / epsH))
    (s : CosmologicalState) (hH : s.H_obs = H_max) :
    dHobs_dt c H_max I_erase epsH s < 0 := by
  unfold dHobs_dt; rw [hH, sub_self, mul_zero, zero_sub]
  exact neg_lt_zero.mpr (mul_pos hIe htanh)

theorem dIdot_dt_nonpos_at_ceiling
    (e f g P_input c gamma_max H_max I_erase epsH epsI : ℝ)
    (he : 0 ≤ e) (hg : 0 < g) (hMH : 0 ≤ M_H c gamma_max H_max I_erase)
    (s : CosmologicalState)
    (hI : s.I_dot = I_dot_max e f g P_input c gamma_max H_max I_erase)
    (hdH : dHobs_dt c H_max I_erase epsH s ≤ M_H c gamma_max H_max I_erase)
    (htanh_nonneg : 0 ≤ Real.tanh (s.I_dot / epsI))
    (htanh_le_one : Real.tanh (s.I_dot / epsI) ≤ 1) :
    dIdot_dt e f g P_input c H_max I_erase epsH epsI s ≤ 0 := by
  unfold dIdot_dt I_dot_max at *
  have hg_ne : g ≠ 0 := ne_of_gt hg
  have h_g_mul : g * s.I_dot = e * M_H c gamma_max H_max I_erase + f * P_input := by
    rw [hI]; exact mul_div_cancel₀ (e * M_H c gamma_max H_max I_erase + f * P_input) hg_ne
  have h_step1 : Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s ≤
      Real.tanh (s.I_dot / epsI) * M_H c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_left hdH htanh_nonneg
  have h_step2 : Real.tanh (s.I_dot / epsI) * M_H c gamma_max H_max I_erase ≤
      1 * M_H c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_right htanh_le_one hMH
  have h_step3 : Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s ≤
      M_H c gamma_max H_max I_erase := by linarith
  have h_step4 : e * (Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s) ≤
      e * M_H c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_left h_step3 he
  have h_assoc : e * Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s =
      e * (Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s) :=
    mul_assoc e (Real.tanh (s.I_dot / epsI)) (dHobs_dt c H_max I_erase epsH s)
  linarith
```

5 Exactness of $(\gamma^*, \dot{I}^*)$ and Multi-Scale Gate Perturbation Bounds

Gating only the erasure sink $\dot{I}_{\text{erase}} \tanh(H_{\text{obs}}/\epsilon_H)$ and the derivative-feedback term $e \tanh(\dot{I}/\epsilon_I)(dH_{\text{obs}}/dt)$—while keeping $-b\gamma$ and $-g\dot{I}$ linear—ensures that $(\gamma^*, \dot{I}^*)$ are exact and the first-order gate derivative vanishes identically at equilibrium:

Proposition 4 (Exactness of $(\gamma^*, \dot{I}^*)$ and Multi-Scale Gate Perturbation Bounds)**.** *At any equilibrium* $(\gamma^*, H^*_{obs}, \dot{I}^*)$ *of Eqs.* (3)–(5)*:*

1. *Since* $dH_{obs}/dt|_* = 0$*, Eq.* (5) *reduces at steady state to* $f\mathcal{P}_{input} - g\dot{I}^* = 0$ *and Eq.* (3) *reduces to* $K(1 - \gamma^*/\gamma_{\max}) - b\gamma^* = 0$ *with* $K \equiv a\mathcal{E}_{avail}\dot{I}^*$*. Thus* $\dot{I}^* = f\mathcal{P}_{input}/g$ *and* $\gamma^* = K\gamma_{\max}/(b\gamma_{\max} + K)$ *are* ***exact*** *(zero* ϵ*-dependence), while* H^*_{obs} *solves* $c\gamma^*(H_{\max} - H^*_{obs}) = \dot{I}_{erase} \tanh(H^*_{obs}/\epsilon_H)$ *and differs from* $H_{\max} - \dot{I}_{erase}/(c\gamma^*)$ *by* $O(e^{-2H^*_{obs}/\epsilon_H})$*.*
2. *When linearizing Eq.* (5) *about the equilibrium, the product rule applied to* $e \tanh(\dot{I}/\epsilon_I)(dH_{obs}/dt)$

yields

$$\frac{\partial}{\partial \dot{I}}\left[e\tanh\left(\frac{\dot{I}}{\epsilon_I}\right)\frac{dH_{obs}}{dt}\right]\Bigg|_* = \frac{e}{\epsilon_I}\operatorname{sech}^2\left(\frac{\dot{I}^*}{\epsilon_I}\right)\underbrace{\frac{dH_{obs}}{dt}\Big|_*}_{=0} + e\tanh\left(\frac{\dot{I}^*}{\epsilon_I}\right)\underbrace{\frac{\partial(dH_{obs}/dt)}{\partial \dot{I}}\Big|_*}_{=0} = 0. \tag{13}$$

Hence the $(3,3)$ *entry of the Jacobian is exactly* $-g$*, with no* $\operatorname{sech}^2(\dot{I}^*/\epsilon_I)$ *linear damping artifact.*

3. *In coordinates* $(\gamma, H_{obs}, \dot{I})$*, the exact Jacobian of the upgraded system* (3)–(5) *at equilibrium is*

$$J_{v1.5} = \begin{pmatrix} -\alpha & 0 & \beta \\ \delta_\epsilon & -\mu_\epsilon & 0 \\ e_{eff}\delta_\epsilon & -e_{eff}\mu_\epsilon & -g \end{pmatrix}, \tag{14}$$

where α *and* β *are identical to v1.1–v1.4,* $e_{eff} = e\tanh(\dot{I}^*/\epsilon_I)$*,* $\delta_\epsilon = c(H_{\max} - H^*_{obs})$*, and* $\mu_\epsilon = c\gamma^* + \frac{\dot{I}_{erase}}{\epsilon_H}\operatorname{sech}^2(H^*_{obs}/\epsilon_H)$*. Two distinct exponential gate scales govern the departure from v1.4 as* $\epsilon_H, \epsilon_I \to 0^+$*:*

- ***Processing gate scale*** (ϵ_I)***:*** $1 - \tanh(\dot{I}^*/\epsilon_I) \sim 2e^{-2\dot{I}^*/\epsilon_I}$*, which dominates* $\|J_{v1.5} - J_{v1.4}\|_F = O(e^{-2\dot{I}^*/\epsilon_I})$ *and* $|e_{c,v1.5} - e_{c,v1.4}|$*, and via the second and third derivatives of* $\tanh(\dot{I}/\epsilon_I)$ *shifts* l_1 *by* $O(\epsilon_I^{-2}\operatorname{sech}^2(\dot{I}^*/\epsilon_I))$*.*
- ***Entropy gate scale*** (ϵ_H)***:*** $\operatorname{sech}^2(H^*_{obs}/\epsilon_H) \sim 4e^{-2H^*_{obs}/\epsilon_H}$*, which shifts* H^*_{obs} *and* μ_ϵ *by* $O(e^{-2H^*_{obs}/\epsilon_H})$*.*

6 Symbolic Tensor Audit, ϵ-Scaling Sweep, and Post-Hopf Benchmark

The script `verification_patch_v1_5.py` executes a five-layer mechanical verification chain across two standard parameter slices from the prior lineage:

- **v1.3/v1.4 Audit Slice (Tables 1 and 2):** $a = 0.5, b = 1, c = 0.5, f = g = 1, \mathcal{E}_{\text{avail}} = 2, \mathcal{P}_{\text{input}} = 1, \dot{I}_{\text{erase}} = 0.5, \gamma_{\max} = 1, H_{\max} = 10$, where $e_c = 135/26 \approx 5.192308$ and $(\gamma^*, H^*_{\text{obs}}, \dot{I}^*) = (0.5, 8.0, 1.0)$.
- **v1.1 Dynamical Benchmark (Table 3):** $a = b = c = f = g = \mathcal{E}_{\text{avail}} = 1, \gamma_{\max} = 10, H_{\max} = 100, \mathcal{P}_{\text{input}} = 5, \dot{I}_{\text{erase}} = 10$, where $e_c = 44.880952/\dot{I}_{\text{erase}} = 4.488095$ and $(\gamma^*, H^*_{\text{obs}}, \dot{I}^*) = (10/3, 97.0, 5.0)$.
- **Verification Chain Summary:** (1) **Lean 4 Kernel** (`boundary_verification.lean`) certifies all 6 face/ceiling conditions of $\mathcal{B}$ (`compact_box_nagumo_certificate`, Scope: Theorem 3); (2) **SymPy Symbolic Audit** verifies $f_1 = f_2 = f_3 = 0$ at steady state and $J_{\text{v1.5}}, B_{\text{v1.5}}, C_{\text{v1.5}}$ (including $C_1 = 0$) with zero symbolic residual (Scope: Proposition 4 tensor construction); (3) **60-Digit `mpmath`** verifies $l_{1,\text{v1.4}}$ against the 50-digit exact reference ($|\Delta| < 10^{-51}$); (4) **750/1450-Digit `mpmath`** certifies the two-gate perturbation bounds and 5-point ϵ-scaling sweep (Tables 1–2); and (5) **`LSODA` + 3-IC Poincaré + Transverse Floquet Audit** certifies post-Hopf confinement, multi-IC convergence ($\max \|\Delta y_{\text{IC}}\|_2 = 1.83 \times 10^{-8} < 10^{-6}$), and orbital asymptotic stability ($\rho(DP) = 8.96 \times 10^{-6} \ll 1$, Table 3).

Table 1: High-precision spectral, Frobenius-norm, and two-gate scale verification at the v1.3/v1.4 audit slice ($\epsilon_H = \epsilon_I = 0.01$, $e_c = 135/26$).

Quantity	**Verified Value (60-Digit / 750-Digit `mpmath` & `SymPy` Audit)**
SymPy equilibrium & tensor audit	Exact zero residual for $f_1 = f_2 = f_3 = 0$ and $J_{\text{v1.5}}, B_{\text{v1.5}}, C_{1,\text{v1.5}} = 0, C_{2,\text{v1.5}}, C_{3,\text{v1.5}}$
Analytic Hopf threshold e_c	$135/26 = 5.192307692307692307692307692\ldots$
Hopf frequency ω_c (s^{-1})	$0.392232270276368063848324 93\ldots$
First Lyapunov coefficient $l_1 = \text{Re}(G_{21})/(2\omega_c)$	$+0.519589117351220268983393 19\ldots$ (subcritical, $l_1 > 0$)
Processing gate deviation $1 - \tanh(\dot{I}^*/\epsilon_I)$	$2.76779305347 \times 10^{-87}$ ($\dot{I}^*/\epsilon_I = 100$)
Entropy gate deviation $\text{sech}^2(H^*_{\text{obs}}/\epsilon_H)$	$5.38132158610 \times 10^{-695}$ ($H^*_{\text{obs}}/\epsilon_H = 800$)
Soft-wall equilibrium shift $\lvert H^*_{\text{obs,v1.5}} - H^*_{\text{obs,v1.4}}\rvert$	$5.38132158610 \times 10^{-695}$ ($< 10^{-694} \ll 10^{-50}$)
Jacobian Frobenius norm $\lVert J_{\text{v1.5}} - J_{\text{v1.4}}\rVert_F$	$1.48135280746 \times 10^{-86}$ ($< 10^{-85} \ll 10^{-50}$)
Hopf threshold shift $\lvert e_{c,\text{v1.5}} - e_{c,\text{v1.4}}\rvert$	$1.43712331623 \times 10^{-86}$ ($< 10^{-85} \ll 10^{-50}$)
First Lyapunov shift $\lvert l_{1,\text{v1.5}} - l_{1,\text{v1.4}}\rvert$	$1.10793084174 \times 10^{-82}$ ($< 10^{-81} \ll 10^{-50}$)
Linear gate derivative $\partial_{\dot{I}}[e\tanh(\dot{I}/\epsilon_I)(dH_{\text{obs}}/dt)]_*$	0 (vanishes identically since $dH_{\text{obs}}/dt\rvert_* = 0$)

Table 2: Parametric gate-width scaling sweep on the v1.3/v1.4 audit slice ($\epsilon_H = \epsilon_I = \epsilon$), demonstrating exponential convergence of the v1.5 soft-wall system to v1.4 as $\epsilon \to 0^+$.

Gate Width ϵ	$1 - \tanh(\dot{I}^*/\epsilon)$	$\lVert J_{\mathbf{v1.5}} - J_{\mathbf{v1.4}}\rVert_F$	$\lvert e_{c,\mathbf{v1.5}} - e_{c,\mathbf{v1.4}}\rvert$	$\lvert l_{1,\mathbf{v1.5}} - l_{1,\mathbf{v1.4}}\rvert$
$\epsilon = 0.100$	4.12231×10^{-9}	2.20630×10^{-8}	2.14043×10^{-8}	1.74819×10^{-6}
$\epsilon = 0.050$	8.49671×10^{-18}	4.54753×10^{-17}	4.41175×10^{-17}	1.39640×10^{-14}
$\epsilon = 0.020$	7.44015×10^{-44}	3.98205×10^{-43}	3.86316×10^{-43}	7.49478×10^{-40}
$\epsilon = 0.010$	2.76779×10^{-87}	1.48135×10^{-86}	1.43712×10^{-86}	1.10793×10^{-82}
$\epsilon = 0.005$	3.83034×10^{-174}	2.05004×10^{-173}	1.98883×10^{-173}	6.11280×10^{-169}

Table 3: Post-Hopf trajectory confinement on the v1.1 benchmark ($\dot{I}_{\text{erase}} = 10, e_c = 4.488095, \epsilon_H = \epsilon_I = 0.05$). At $e = 1.20\, e_c$, a Poincaré return map on $\Sigma = \{\gamma = 1.0\ \text{s}^{-1}, \dot{\gamma} > 0\}$ confirms convergence across 3 widely separated initial conditions in $\mathcal{B}$ ($\max \lVert \Delta y_{\text{IC}}\rVert_2 = 1.83 \times 10^{-8} < 10^{-6}$) to an attracting periodic relaxation orbit at $(H_{\text{obs}}, \dot{I})_\Sigma = (3.91083047, 21.68392737)$ with period $T = 18.95704705$ s, 5-cycle period spread $< 10^{-6}$ s (2.58×10^{-9} s on x86_64 Linux; 1.72×10^{-7} s on ARM64), state spread $\lVert \Delta y\rVert_2 < 10^{-6}$ (3.85×10^{-10} on x86_64; 5.57×10^{-9} on ARM64), and transverse Floquet spectral radius $\rho(DP) = 8.96 \times 10^{-6} \ll 1$.

Gain	**v1.0 / v1.1 Behaviour**	**v1.5** $\min \gamma(t)$	**v1.5** $\max \gamma(t)$	**v1.5** $\min H_{\text{obs}}(t)$	**v1.5 Observed Regime in** $\mathcal{B}$ ($t \in [0, 400]$)
$e = 0.90\, e_c$	Decays to eq. (6)	3.333272	3.333433	96.999947	Asymptotically stable equilibrium
$e = 1.01\, e_c$	Bounded osc. (5.4×10^{-3})	3.331644	3.334981	96.998534	Bounded interior oscillation
$e = 1.04\, e_c$	Escapes: $\gamma \to 0^-$ at $t \approx 259$	0.005682	9.720486	0.003339	Attracting relaxation cycle in $\mathcal{B}$
$e = 1.08\, e_c$	Escapes: $\gamma \to 0^-$ at $t \approx 136$	0.005461	9.730970	0.003222	Attracting relaxation cycle in $\mathcal{B}$
$e = 1.20\, e_c$	Escapes: $\gamma \to 0^-$ at $t \approx 55$	0.004889	9.758112	0.002921	Attracting cycle ($T = 18.957047$ s, $\rho = 8.96 \times 10^{-6}$)
$e = 2.00\, e_c$	Escapes: $\gamma \to 0^-$ at $t \approx 19$	0.002885	9.854966	0.001854	Attracting relaxation cycle in $\mathcal{B}$

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